VOCABULARY FLASH CARD

A VOCABULARY LEARNING WEBSITE USING HTML AND CSS

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BONAFIDE CERTIFICATE

Certified that this project report titled **VOCABULARY FLASH CARD: A VOCABULARY LEARNING WEBSITE USING HTML AND CSS** is the bonafide work of **REEMATHASLIM S (241001193)** who carried out the work during her Mini-Project work. Certified further that based on the task given, she has completed this project under the supervision of the faculty.

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ABSTRACT

Vocabulary Flash Card is a simple educational website designed to help learners practise and improve their English vocabulary. The website presents words as digital flash cards. Each card contains a vocabulary word and its meaning, making it easy for users to revise common words in a clear and organised way.

The project is developed using only HTML5 and CSS3. HTML is used to create the page structure, headings, card content and navigation links. CSS is used to design the layout, colours, typography, spacing and card appearance. The interface is designed to be simple, readable and suitable for students who want to learn new words during short study sessions.

The flash cards are grouped into a collection of useful vocabulary items. Each card displays a word, its meaning and a sample sentence to show how the word can be used in context. CSS hover effects provide visual feedback when a user moves the pointer over a card. The project does not use JavaScript, a database or a backend service; the learning content is presented as static webpage content.

This mini-project demonstrates the use of basic web development concepts, semantic HTML elements and CSS styling to create an educational resource that can be opened in a modern web browser.

1. INTRODUCTION

1.1 Overview

Vocabulary is an important part of language learning. Knowing the meaning of words helps learners understand reading materials, communicate clearly and express ideas with confidence. However, remembering unfamiliar words can be difficult when they are presented only as long lists.

A flash card is a learning aid that places a prompt on one side and related information on the other. In this project, a vocabulary flash card webpage displays each selected word together with its meaning and an example sentence. The cards are arranged in a clean grid so that learners can scan and revise them easily.

1.2 Problem Statement

Students may find it difficult to revise vocabulary from notebooks or unstructured word lists. Such methods can make it harder to locate a word, understand its usage and maintain interest during revision. A simple, organised webpage can present vocabulary in a more attractive and accessible format. **1.3 Objectives**

- To create a simple website for learning English vocabulary.
- To present words, meanings and example sentences using flash-card layouts.
- To use HTML for structured content and CSS for visual design.
- To provide a readable layout that adapts to different screen sizes.
- To make the project easy to open and use in a web browser.

2. LITERATURE REVIEW AND BACKGROUND STUDY

2.1 Vocabulary Learning

Vocabulary learning involves recognising words, understanding their meanings and applying them correctly in sentences. Repeated revision and examples help learners connect a word with its meaning and usage. Flash cards are commonly used as a compact revision method because each item focuses attention on one word or concept.

2.2 Digital Learning Materials

Digital learning materials can organise information in a consistent format and make it available through computers and mobile devices. A webpage can combine text, layout and visual emphasis without requiring printed materials. This project applies that idea to a small vocabulary collection.

2.3 Role of HTML

HTML (HyperText Markup Language) provides the structure of a webpage. Elements such as headings, paragraphs, sections, navigation and article containers organise the learning content. Semantic elements make the page easier to understand and maintain.

2.4 Role of CSS

CSS (Cascading Style Sheets) controls the presentation of HTML content. It is used to define colours, fonts, borders, spacing, card dimensions, grid layouts and responsive behaviour. CSS transitions and hover effects can provide simple visual feedback.

2.5 Summary

The background study supports the use of a structured webpage and card-based presentation for vocabulary revision. HTML and CSS are sufficient to build the static learning interface developed in this project.

3. SYSTEM REQUIREMENTS

3.1 Hardware Requirements

- Desktop computer, laptop, tablet or smartphone.
- Minimum 4 GB RAM recommended for comfortable browser use.
- Standard display and keyboard or touch input.

3.2 Software Requirements

- Operating system: Windows, Linux, macOS or Android.
- Modern web browser such as Google Chrome, Microsoft Edge or Mozilla Firefox.
- Code editor such as Visual Studio Code.
- HTML5 and CSS3 support in the browser.

3.3 Functional Requirements

- Display the project title and introductory information.
- Present vocabulary cards in an organised grid.
- Show each word with its meaning and example sentence.
- Provide clear visual styling for headings and cards.
- Allow users to navigate and read the vocabulary content.

3.4 Non-Functional Requirements

- Usability: simple and clear interface.
- Readability: suitable font sizes, spacing and contrast.
- Responsiveness: layout adjusts to common screen sizes.

- Portability: runs in a browser without installation.
- Maintainability: HTML content and CSS styling are kept separate.

4. SYSTEM DESIGN AND METHODOLOGY

4.1 System Overview

The application follows a simple front-end structure. The browser reads the HTML document and applies the linked CSS stylesheet. The HTML contains the title, navigation, introduction and vocabulary cards. The CSS file controls the visual presentation and responsive layout.

4.2 System Architecture

User → opens the webpage in a browser → **HTML structure** provides the content → **CSS stylesheet** formats the page → **Vocabulary flash cards** are displayed to the user.

4.3 Page Structure

- Header: displays the project name and navigation.
- Introduction section: explains the purpose of the website.
- Vocabulary section: contains the collection of word cards.
- Flash card: contains a word, meaning and sample sentence.
- Footer: displays project or learning-related information.

4.4 Design Approach

The interface uses a consistent colour palette, clear typography and card spacing. A grid layout places cards in columns on wider screens and reduces the number of columns on smaller screens. Rounded corners, subtle shadows and hover transitions make the cards visually distinct without distracting from the text.

4.5 Workflow

- Plan the page sections and vocabulary content.
- Create the HTML document and add semantic elements.
- Create the CSS stylesheet and define the page layout.
- Add vocabulary entries as individual card elements.
- Open the page in a browser and verify the layout.

5. TECHNOLOGY STACK

5.1 HTML5

HTML5 is used to create the structure of the Vocabulary Flash Card website. Common elements include html, head, title, body, header, nav, main, section, article, h1, h2, p and footer. The article element can be used for individual vocabulary cards because each card represents a separate content item.

5.2 CSS3

CSS3 is used to style the webpage. Selectors target HTML elements and classes. Properties such as background-color, color, font-family, padding, margin, border-radius and box-shadow define the appearance. CSS Grid or Flexbox can be used to arrange the cards.

5.3 Responsive Design

Media queries allow the card layout to change according to screen width. For example, a desktop layout may show three cards in a row, while a smaller screen may show two or one card per row. This improves readability on mobile devices.

5.4 Development Tools

- Visual Studio Code — writing and editing HTML and CSS files.
- Web browser — viewing and testing the webpage.
- Developer Tools — checking layout, spacing and responsive behaviour.

5.5 Technologies Not Used

The project uses HTML and CSS only. JavaScript, React, Bootstrap, databases and server-side programming are not part of the implementation. The vocabulary content is static and is written directly in the HTML document.

6. IMPLEMENTATION

6.1 HTML Page Creation

The HTML document begins with the HTML5 doctype declaration. The head section contains the character encoding, viewport setting, page title and link to the external CSS file. The body contains the visible page content.

6.2 Header and Introduction

The header displays the website name. A navigation area can contain links to the Home, Vocabulary and About sections. The introduction briefly explains how the flash cards can be used for vocabulary revision.

6.3 Vocabulary Card Structure

Each vocabulary card is created using a container such as an article element. A heading displays the word, a paragraph gives its meaning and another paragraph provides an example sentence. Repeating this structure keeps all cards consistent.

Example HTML structure:

```
<article class="card">
  <h2>Curious</h2>
  <p>Meaning: eager to know or learn.</p>
  <p>Example: She was curious about the new topic.</p>
</article>
```

6.4 CSS Styling

The stylesheet defines the background, text colour, font, card width, spacing and border radius. Grid or Flexbox is used to align the cards. A hover selector can slightly raise a card or add a shadow when the pointer is placed over it.

6.5 Content Organisation

Vocabulary items are selected to include useful everyday words. Each entry includes a short definition and a sentence that demonstrates the word in context. This makes the card more useful than displaying the word alone.

7. RESULTS AND DISCUSSION

7.1 Expected Output

The completed webpage displays the project title, a short introduction and a collection of vocabulary flash cards. The cards follow a consistent design and present the word, meaning and example sentence in a readable order.

7.2 User Interface

The interface is designed to avoid unnecessary complexity. A clear heading introduces the page, while the card grid makes it possible to view multiple vocabulary items together. The use of spacing and visual grouping helps users identify each word card.

7.3 Responsive Behaviour

The layout is designed to adjust to different screen widths through CSS. On a desktop, multiple cards can appear in one row. On a tablet or mobile screen, the number of columns is reduced to prevent the content from becoming too narrow.

7.4 Project Outcome

- A working static vocabulary learning webpage was created.
- Vocabulary words are presented with meanings and example sentences.

- HTML provides the content structure and CSS provides the visual design.
- The project can be opened directly in a modern browser.
- The implementation does not require a backend or database.

7.5 Limitations

Since the project uses only HTML and CSS, it does not provide automatic card flipping, search, progress tracking, random word selection or dynamic content updates. The vocabulary items are fixed in the HTML source. These features may be considered in a future version using additional technologies.

8. CHALLENGES AND ISSUES FACED

8.1 Organising the Content

One challenge was arranging the word, definition and example sentence so that each card remained easy to read. A consistent HTML structure and spacing rules were used to maintain uniformity.

8.2 Card Alignment

Cards can appear uneven when their content lengths differ. CSS Grid and consistent padding help maintain alignment and create a balanced layout.

8.3 Responsive Layout

A layout that looks suitable on a laptop may not fit a mobile screen. Media queries and flexible card widths help the page adapt to smaller screens.

9. TESTING

The webpage is checked manually in a browser to confirm that the HTML content and CSS styling are displayed correctly.

No.	Test case	Expected result
1	Open the HTML file	Home page loads
2	Check page title	Correct project title appears
3	View vocabulary cards	Words and meanings are visible
4	Check example sentences	Text is readable
5	Resize browser window	Cards adjust to screen width
6	Check navigation links	Links move to the correct section
7	Check CSS file link	Styles are applied correctly

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

The Vocabulary Flash Card mini-project demonstrates how HTML and CSS can be used to create a simple educational website. The webpage organises vocabulary words into individual cards containing definitions and example sentences. The structured layout makes the content easy to read and revise.

The project helped apply fundamental web programming concepts, including HTML document structure, semantic elements, CSS selectors, spacing, typography, card layouts and responsive design. It also demonstrates how a useful learning resource can be created without using a complex framework or backend. **10.2 Future Scope**

- Add JavaScript to flip cards and reveal meanings interactively.
- Add search and category filters for vocabulary words.
- Include audio pronunciation for each word.
- Add quizzes and progress tracking.
- Store user learning progress using browser storage or a database.
- Allow users to add and manage their own vocabulary lists.

10.3 Final Note

The current version focuses on a clear and attractive static interface using only HTML and CSS. The proposed future enhancements can extend the project into a more interactive vocabulary learning application.

11. REFERENCES

- MDN Web Docs — HTML: HyperText Markup Language reference.
- MDN Web Docs — CSS: Cascading Style Sheets reference.
- W3Schools — HTML Tutorial and CSS Tutorial (for basic syntax reference).
- W3C — HTML and CSS standards and specifications.

12. SYSTEM USAGE / USER MANUAL

12.1 Running the Application

- Open the project folder in Visual Studio Code or another text editor.
- Locate the main HTML file (for example, index.html).
- Open the file in a modern web browser, or use the editor's Live Server extension.
- Scroll through the page to view the vocabulary cards.
- Resize the browser window to check the responsive layout.

12.2 User Instructions

Read the vocabulary word shown on each card. Read its meaning and example sentence to understand how it is used. Continue through the remaining cards for revision. The current static version displays the information directly on the page.

GLOSSARY

HTML: HyperText Markup Language, used to structure webpage content.
CSS: Cascading Style Sheets, used to style webpage elements.
Flash Card: A compact learning card containing a prompt and related information.
Responsive Design: A layout approach that adapts to different screen sizes.
Browser: Software used to open and view webpages.